

25PC2RA202: Sensors and Actuators Laboratory (S&A Lab)

Experiment 11: Brushless DC Motor Speed Control

Prof. A. Adithya Kashyap, Prof. Harshvardhan Uppaluru

Date: September 21/October 15, 2026

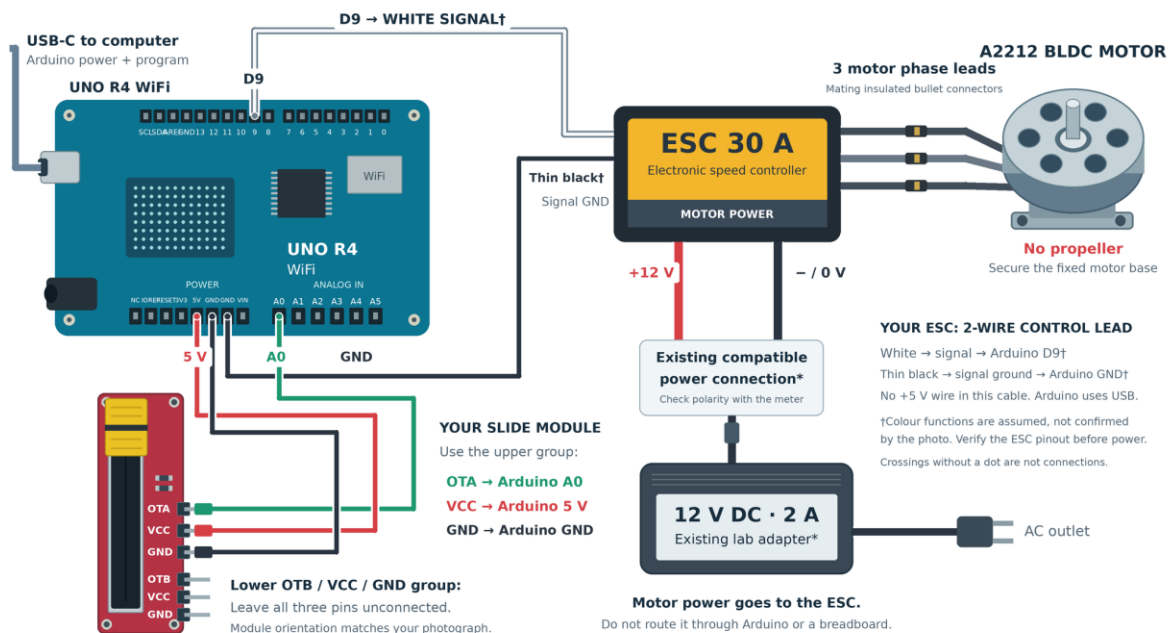
1. Introduction

An electronic speed controller (ESC) connects the Arduino control signal to a brushless DC (BLDC) motor. A linear potentiometer provides the input used to vary the motor speed. The control signal uses pulses of 1000–2000 microseconds, repeated at approximately 50 Hz. The Servo library generates these pulses; the ESC controls the motor power. In this experiment, you must assemble the circuit, run the Arduino code and compare the motor response at different potentiometer positions. Record what you observe rather than treating the command as an RPM measurement.

2. Equipment & Wiring

BLDC speed control — connection picture

UNO R4 WiFi • 10 kΩ slide potentiometer • 30 A ESC • A2212 motor



BEFORE CONNECTING POWER

Disconnect USB and motor power while wiring. Start with the slider low. Keep the motor firmly mounted.

*Use the 12 V, 2 A adapter only if the ESC accepts 12 V, the existing connection is secure and startup is stable.

If that connection is unavailable or the adapter trips, do not power the motor. Do not insert bare wires into its plug.

Pin reference: Arduino UNO R4 WiFi official pinout. Circuit approach: How To Mechatronics ESC/BLDC tutorial.

3. Arduino Code

```
1 // Brushless DC motor speed control
2 // Tutorial: Dejan, How To Mechatronics [1]
3
4 #include <Servo.h>
5
6 Servo motorESC;
7 const int escPin = 9;
8 const int potPin = A0;
9
10 void setup()
11 {
12     // Set output pin and pulse limits (us).
13     motorESC.attach(escPin, 1000, 2000);
14 }
15
16 void loop()
17 {
18     // Read the potentiometer input.
19     int potReading = analogRead(potPin);
20
21     // Convert 0-1023 to the command range.
22     int throttle = map(
23         potReading, 0, 1023, 0, 180);
24
25     // Apply the new command to the ESC.
26     motorESC.write(throttle);
27 }
```

4. Key Concepts - [Link](#)

Using the linked tutorial, study the complete tutorial and especially the following concepts:

- Working principles of BLDC motor
- ESC, Battery Eliminator Circuit (BEC) and motor phases.
- KV rating, servo-style pulses, arming and ESC calibration.

5. Observations

Write the answers to the following questions in your record:

- a) What did you hear when the ESC was powered at minimum throttle? Record the sequence you observed.
- b) At approximately which linear potentiometer position did the motor first rotate? Was there a region with no visible rotation?
- c) How did the motor respond as you slowly moved the potentiometer towards a higher command?
- d) What happened when the potentiometer returned to minimum? Record any delay before the rotor stopped.

- e) For analog input values 0, 512 and 1023, calculate the mapped command produced by the sketch.
- f) What does 180 mean in this program? Does it instruct the BLDC motor to turn through 180 degrees?
- g) Identify the control-signal wire, the battery wires and the three motor wires in your circuit.

6. Evaluation Rubric

Assessment Component	Marks	Awarded Marks
Wiring	3	
Programming	2	
Observations	4	
Attendance	1	
Total	10	

Signatures

Prof. A. Adithya Kashyap
Date:

Prof. Harshvardhan Uppaluru
Date: